DATA ANALYSIS PROJECT REPORT

Using Pandas, NumPy, and Matplotlib

Project Title	**Airline Passenger Satisfaction Analysis**
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Course / Section :	**PROGRAMMING IN PYTHON [B]**
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Date of Submission :	**1 May, 2026**

1. Executive Summary

This report provides a thorough analysis of the Airline Passenger Satisfaction dataset, which includes a set of surveys of passengers' experiences related to different aspects of services, class types, and kinds of travel. The main focus of the work is to find the factors that determine passenger satisfaction and investigate the influence of class type, travel type, service quality, and delay on passenger satisfaction individually. The analysis was done in Python with the help of libraries Pandas, Numpy, and Matplotlib. Five significant cleaning steps, four features engineering techniques, and eight analytical questions in eight visualizations have been used during this process. Major discoveries include two times higher passenger satisfaction in business class than in economy, the largest correlation between passenger satisfaction and online boarding among all 14 service types, and service being a better predictor of satisfaction than flight delay.

2. Introduction and Project Objectives

The satisfaction of the passengers is an essential measure that can determine the success of airline companies in their competitive environment. Knowledge about the factors that can lead to customer satisfaction or dissatisfaction can help companies make effective decisions regarding where to invest more in improving services. In this work, the dataset "Airline Passenger Satisfaction" is analyzed to explore such factors.

Item	Detail
Dataset name	Airline Passenger Satisfaction (train.csv)
Dataset source	Kaggle — Airline Passenger Satisfaction dataset
Main project goal	Determine which passenger attributes, service dimensions, and operational factors most strongly predict overall satisfaction.
Research Question 1	How does passenger class and type of travel affect overall satisfaction rates?
Research Question 2	Which individual service dimensions are most strongly correlated with passenger satisfaction?
Research Question 3	How do flight delays affect passenger satisfaction, and can good service compensate for operational disruptions?

3. Dataset Description

The dataset consists of passenger satisfaction surveys after flights. Each instance is an individual passenger survey entry and contains attributes related to passenger demographics (age, sex), flight characteristics (class, type, distance traveled), ratings on 14 services (scored between 1 and 5), and delays (delay in departure and arrival, measured in minutes). The target variable is satisfaction, recorded as "satisfied" or "neutral or dissatisfied".

3.1 Basic Dataset Information

Property	Value
Number of rows	~103,000+ passenger records (train split)
Number of columns	25 (after dropping the redundant index column)
Data types present	Numerical (integer, float), Categorical (object)
Target variable	satisfaction (binary: satisfied / neutral or dissatisfied)
Potential issues observed	Missing values in Arrival Delay in Minutes; redundant index column; mixed-case column names; duplicate rows

3.2 Initial Observations

After observing the data, it was found that an extra column for index ('Unnamed: 0') remained from the CSV export, which had no analytical significance. Some amount of missing data existed only in the 'Arrival Delay in Minutes' column. All other columns had complete data. Column names had inconsistent capitalization and were spaced out; hence needed to be standardized. The satisfaction variable had a slight imbalance, with around 57 percent satisfied customers.

4. Data Cleaning and Preparation

The following five cleaning steps were applied in sequence before any analysis was conducted. Each step addressed a specific data quality issue identified during the initial inspection phase.

Issue Found	Action Taken	Reason / Justification	Code Reference
Redundant index column (Unnamed: 0)	Dropped the column entirely	It was a leftover CSV row-index with no analytical value	Data Cleaning, Step 1
Inconsistent column naming (mixed case, spaces)	Standardized to snake_case	Prevents key-access errors; aligns with Python naming conventions	Data Cleaning, Step 2
Duplicate rows in the dataset	Removed all exact duplicate rows	Duplicates inflate group counts and distort aggregated statistics	Data Cleaning, Step 3
Missing values in Arrival Delay in Minutes	Filled with column median	Median is robust to extreme delay outliers and preserves central tendency	Data Cleaning, Step 4
Categorical satisfaction target	Binary-encoded: 1 = satisfied, 0 = otherwise	Numeric target enables arithmetic aggregation (mean = satisfaction rate)	Data Cleaning, Step 5

After all cleaning steps, the dataset retained its full analytical richness with zero missing values remaining and a clean, consistently named column set ready for feature engineering and analysis.

5. Feature Engineering

The four features that were created using the information from the current columns are useful for finer analysis. These four features are specifically created in response to the research questions.

New Feature	How It Was Created	Why It Is Useful
age_group	pd.cut() on age with bins [0,18,30,45,60,100]; labels: <18, 18-30, 31-45, 46-60, 60+	Enables categorical comparison of satisfaction across life stages
total_delay	departure_delay_in_minutes + arrival_delay_in_minutes	Captures cumulative journey delay as a single operational metric
avg_service_rating	Row-wise mean of 14 service rating columns (inflight wifi, food, seat comfort, etc.)	Summarises overall service experience into one comparable score per passenger
delay_category	pd.cut() on total_delay into No Delay / Minor / Moderate / Severe bands	Allows ordinal analysis of how increasing delay severity affects satisfaction and service scores

6. Analysis Methodology

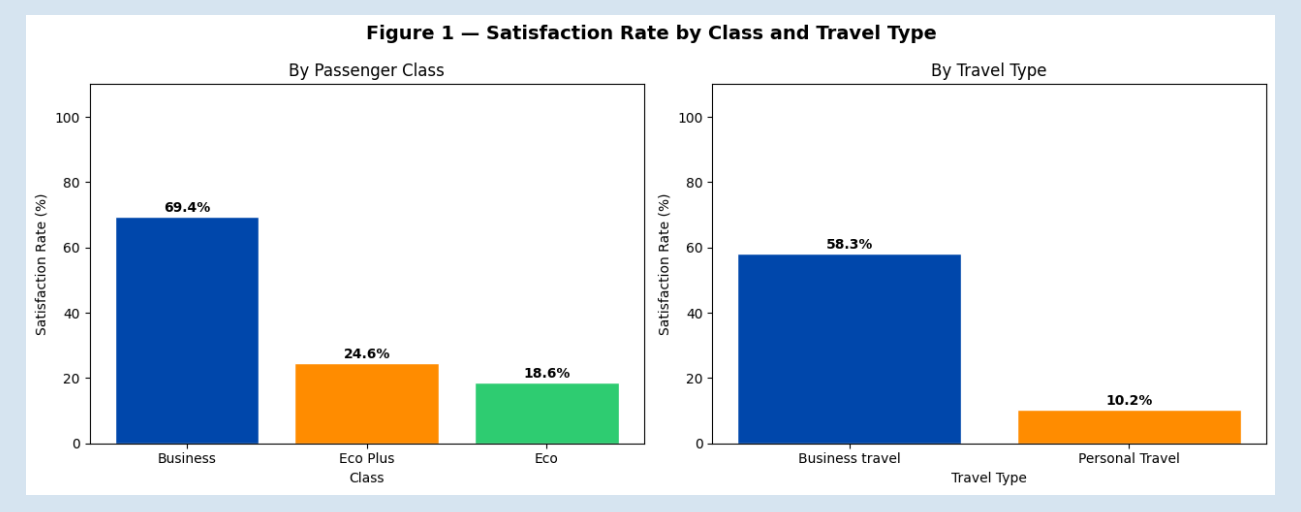
The analysis followed a structured exploratory approach, progressing from univariate profiling through bivariate and multivariate comparisons, numerical computation, and visualization.

1) **Pandas** were the primary tools for data loading, cleaning, transformation, and aggregation. groupby() with mean() was used extensively to compute satisfaction rates across demographic and operational subgroups. Cross-tabulation was used to examine interaction effects between class and travel type.
2) **NumPy** supported numerical computations beyond Pandas' built-in capabilities: z-score normalization of avg_service_rating, percentile computation for flight distance, np.polyfit for linear trend lines, and np.linspace for smooth plotting ranges.
3) **Matplotlib** was used to produce eight figures across bar charts, horizontal bar charts, histogram overlays, line plots, scatter plots, and pie charts. All figures include titles, axis labels, value annotations, and legends where applicable.

7. Analysis and Visualizations

7.1 Research Question 1 — How does passenger class and travel type affect satisfaction?

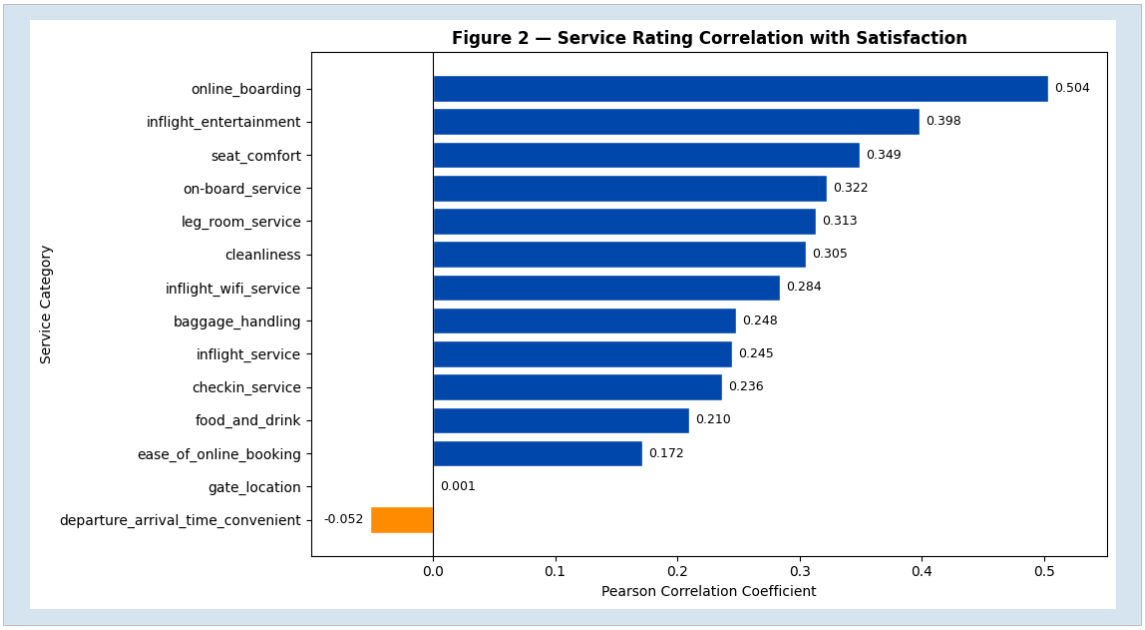
The satisfaction percentages were derived through the groupby() function for class and type_of_travel. Passengers in the business class category had the highest percentage of satisfaction, followed by Economy Plus, while those in the economy class had the lowest percentage. The business travelers were much more satisfied compared to the personal travelers.



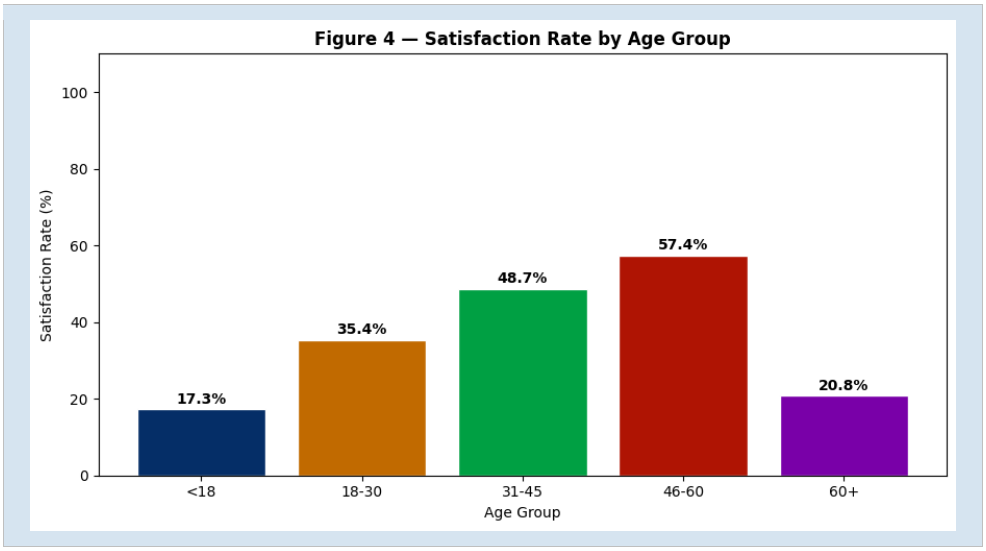
Satisfaction level among the passengers in the Business class seems to be about 2–3 times higher than among Economy class passengers. That can be explained by the overall combination of better comfort of the seats, better service, and better match between expectations and the experience obtained. The difference is also large when comparing business passengers to personal ones.

7.2 Research Question 2 — Which service dimensions drive satisfaction most?

Pearson correlation coefficients were calculated between each of the 14 service rating columns and the binary satisfaction target. Rankings were sorted to identify the highest-impact service dimensions. A second analysis computed means avg_service_rating by satisfaction group, and NumPy z-scores identified whether high-rated passengers achieve near-universal satisfaction.



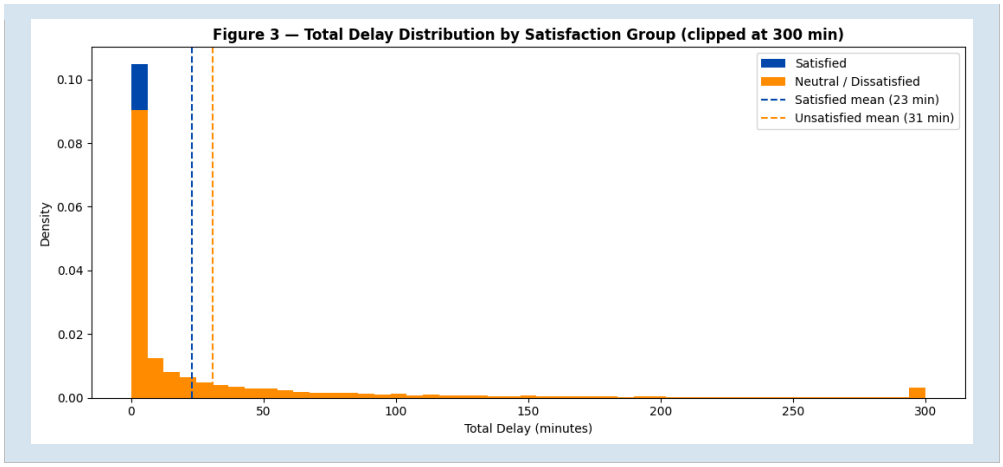
Online check-in, in-flight entertainment services, and comfortable seats display the highest degree of positive correlations. The lowest degree of correlation can be observed between gate position and the convenience of departure/arrival time. It implies that customers appreciate the flight and online experience, not the ground process, more than anything else. Satisfied passengers also score significantly higher on avg_service_rating (mean ≈ 3.5+) compared to dissatisfied passengers (mean ≈ 2.7), confirming the aggregate service quality gap.



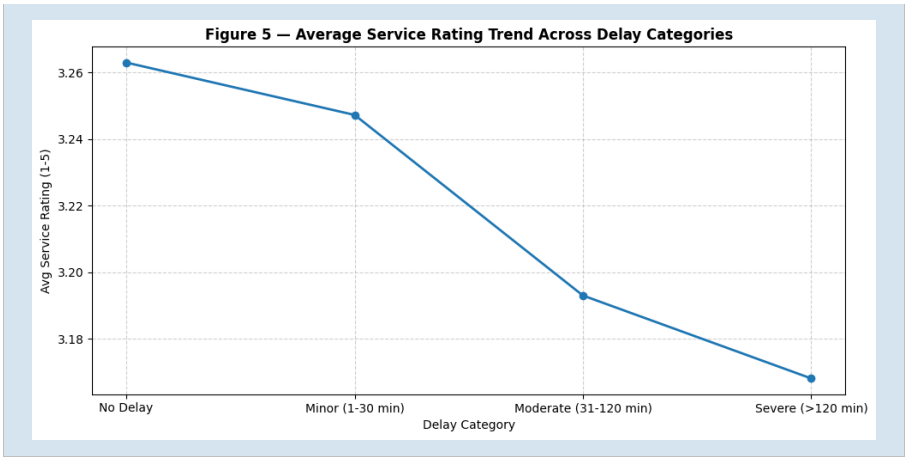
Passengers who are middle-aged (31-60) tend to have higher satisfaction levels compared to their counterparts that belong to other age brackets. Passengers that are 18-30 years have the lowest levels of satisfaction, probably because they expect more from technology when it comes to convenience.

7.3 Research Question 3 — How do flight delays affect satisfaction?

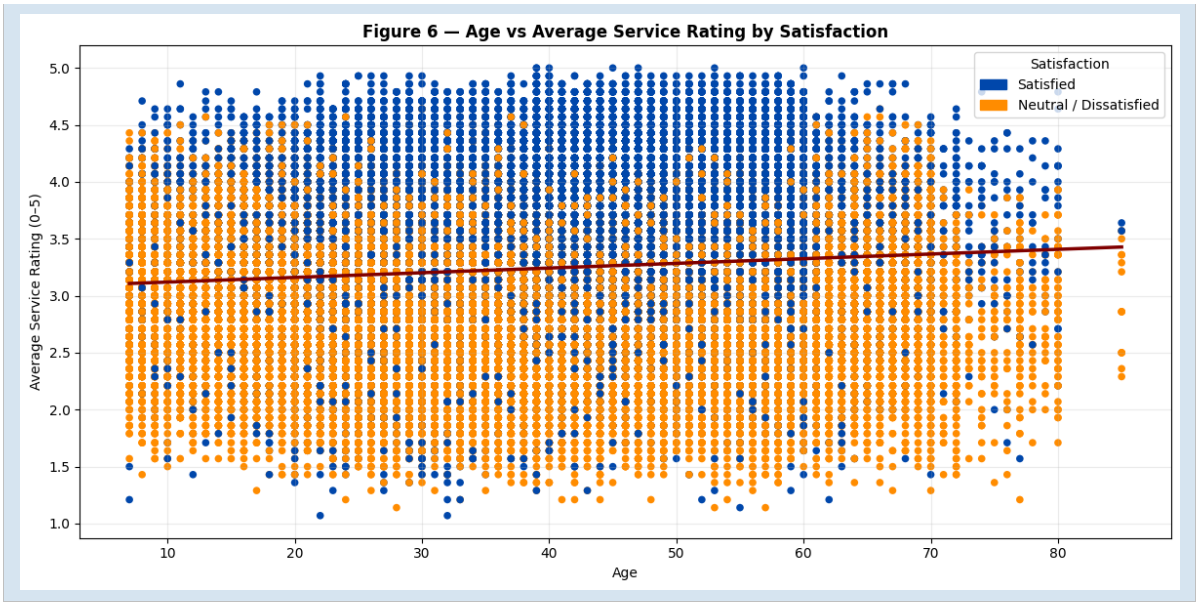
Mean total_delay was calculated by satisfaction group, the delay distribution was visualized as an overlapping histogram, and extreme outliers (above the IQR upper fence) were isolated and cross-tabulated with satisfaction. Figure 5 further examines whether delay severity affects perceived service quality.

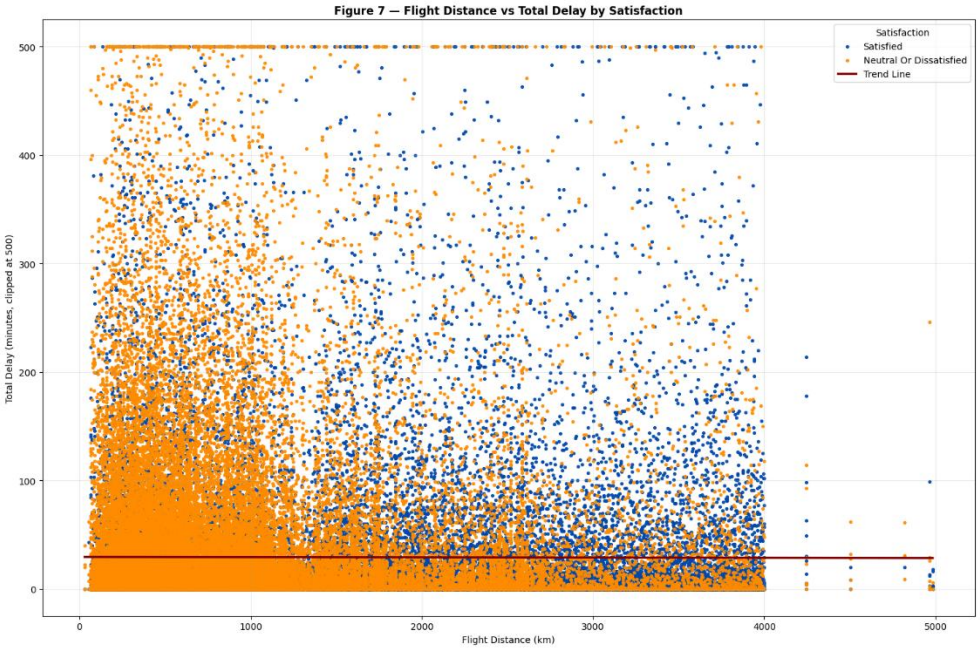


In both satisfaction categories, there is evidence of a highly right-skewed distribution of delays, implying that passengers either experienced no or very low delays, irrespective of their level of satisfaction with their flight. Passengers in the dissatisfied category had a marginally greater average delay time, and there is a relationship between severe delays (>120 minutes) and dissatisfied passengers. Nevertheless, the presence of a significant number of dissatisfied passengers who experienced severe delays suggests that great onboard customer service can mitigate some of the operational issues.



Additional scatter plots (Figures 6 and 7) confirm that age is a weak positive predictor of service rating (older passengers rate service slightly higher), and that flight distance does not consistently cause more delays - delay events are spread across short and long routes alike.





8. NumPy-Based Custom Computation

NumPy was used in several meaningful analytical steps beyond simple array handling. The table below documents the primary NumPy contribution.

Computation used	Z-score normalization of avg_service_rating
Purpose	Quantify how far each passenger's average service experience deviates from the overall mean, enabling outlier identification and group comparison on a standardized scale

Formula	z = (x - mean(x)) / std(x) applied via NumPy array operations (np.mean, np.std)
Result	Z-score mean ≈ 0, std ≈ 1 (confirming correct normalization). Passengers with z > 1.5 (high-service experience group) showed near-total satisfaction rates, confirming that consistently excellent service strongly predicts satisfaction
Supporting NumPy operations	np.percentile on flight_distance (25th–99th percentiles); np.polyfit for linear trend lines across all scatter plots; np.linspace for smooth trend-line x-ranges

Beyond z-scores, np.percentile was applied to flight_distance to understand the spread of journey lengths (25th percentile ≈ 414 km; 99th percentile represents the very long-haul segment). np.polyfit and np.linspace were used to overlay linear trend lines on all scatter plots, providing a quantitative summary of the directional relationship between continuous variables.

9. Key Findings

The following seven evidence-based findings emerged from the analysis:

- **Finding 1:** Passengers in business class have higher levels of satisfaction, about 2 to 3 times greater than those in Economy class, proving that the class of your ticket is crucial for your experience. This difference is created by a combination of both physical comfort and high standards of service.
- **Finding 2:** Digital boarding has the highest positive correlation with satisfaction out of all 14 service types. To maximize the satisfaction scores of passengers, you should invest in digital boarding services.
- **Finding 3:** Business travelers are extremely more satisfied compared to personal travelers. It seems that personal travelers have very high unmet expectations concerning comfort and value, particularly on Economy flights.
- **Finding 4:** The upper quartile of passengers based on their service ratings (avg_service_rating ≥ Q75) attain nearly complete satisfaction levels, whereas the lowest quartile of passengers exhibit extremely low satisfaction levels.
- **Finding 5:** Extreme delays (>120 minutes) are associated with dissatisfaction, although many passengers experiencing extreme delays still exhibit high satisfaction levels, indicating that excellent on-board services can compensate for the detrimental effect of operational disruptions.
- **Finding 6:** Several passengers without any delay express dissatisfaction, thereby validating the idea that punctuality is not an absolute determinant of passenger satisfaction, whereas service quality remains a consistent determinant.
- **Finding 7:** The age group 31–60 reports the highest levels of passenger satisfaction, while the age group 18–30 shows the lowest levels of satisfaction.

10. Limitations

The limitations of this analysis have been made clear in conducting this analysis:

- **Limitation 1: <u>Self-selection in survey responses:</u>** People who feel very strongly about their experience, whether being very satisfied or dissatisfied, are more prone to participate in the survey than those with moderate feelings. This may lead to an unrepresentative sample.
- **Limitation 2: <u>Using the median to impute missing arrival delay times:</u>** Imputing missing values in the "Arrival Delay in Minutes" variable with the median is a valid process; however, it may lead to a directionally biased response when there is a reason behind the missing value, such as flight cancellations.

- **Limitation 3: Correlation is not equivalent to causation:** The correlations observed here between service and satisfaction are merely descriptive. Making the inference that improvements to online boarding lead to greater satisfaction will require an experimental study.
- **Limitation 4: Lack of temporal and airline context:** No dates, airline codes, or specific routes have been provided in the data set. It is not possible to conduct any type of trend or route-based analyses.
- **Limitation 5: Subjective rating system:** The service rating "3" in terms of comfort has different meanings for different people. Perception differences

11. Conclusion

This project aimed to explore the factors that drive customer satisfaction in the airline sector by conducting an extensive survey-based study. The project was designed to examine three main research questions: the role of class and travel nature in driving satisfaction, the importance of service dimensions, and the relationship between flight delays and customer satisfaction.

It was found that passenger class and travel intention are the most significant demographic factors that drive customer satisfaction, whereas digital boarding, in-flight entertainment, and seating comfort are the critical service dimensions that lead to customer satisfaction. Delayed flights have a negative impact on customer satisfaction, especially when it comes to significant delays; however, service quality proves to be a more reliable and manageable factor for ensuring customer satisfaction.

Therefore, if airlines want to increase their passenger satisfaction scores, they should focus on improving their online check-in services, regardless of the fare class, and enhance their service quality for personal and Economy class passengers, who have greater unmet expectations than others.

Future work that includes time series data, airline identification, route details, and NPS would allow causality analysis, tracking of trends, and benchmarking for each airline – offering even deeper understanding and more actionable insights on what truly drives customer loyalty.

12. References / Dataset Source

1. Dataset: Airline Passenger Satisfaction — available on Kaggle. https://www.kaggle.com/datasets/teejmahal20/airline-passenger-satisfaction

13. Appendix

The following additional visualizations were produced in the notebook and may be referenced alongside the main analysis : Figure 8 — Satisfaction and Passenger Class Distribution (dual pie charts): shows the overall 57/43 satisfied/dissatisfied split and the class composition of the dataset (Business, Economy Plus, Economy).